

EMPLOYEE MANAGEMENT SYSTEM



CSE 200: Software Development Project II

SUBMITTED BY

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Abstract

This project is an Employee Management System developed using Python (Tkinter) and MySQL. The system allows users to manage employee records efficiently including adding, updating, deleting, and searching employee information. The application provides a user-friendly graphical interface along with a secure login system. The main goal of this project is to simplify employee data handling and improve productivity.

Chapter 1: Introduction

1.1 Problem Specification

Managing employee data manually is time-consuming and error-prone. Organizations need an efficient digital system to handle employee records such as name, email, department, and salary.

1.2 Objectives

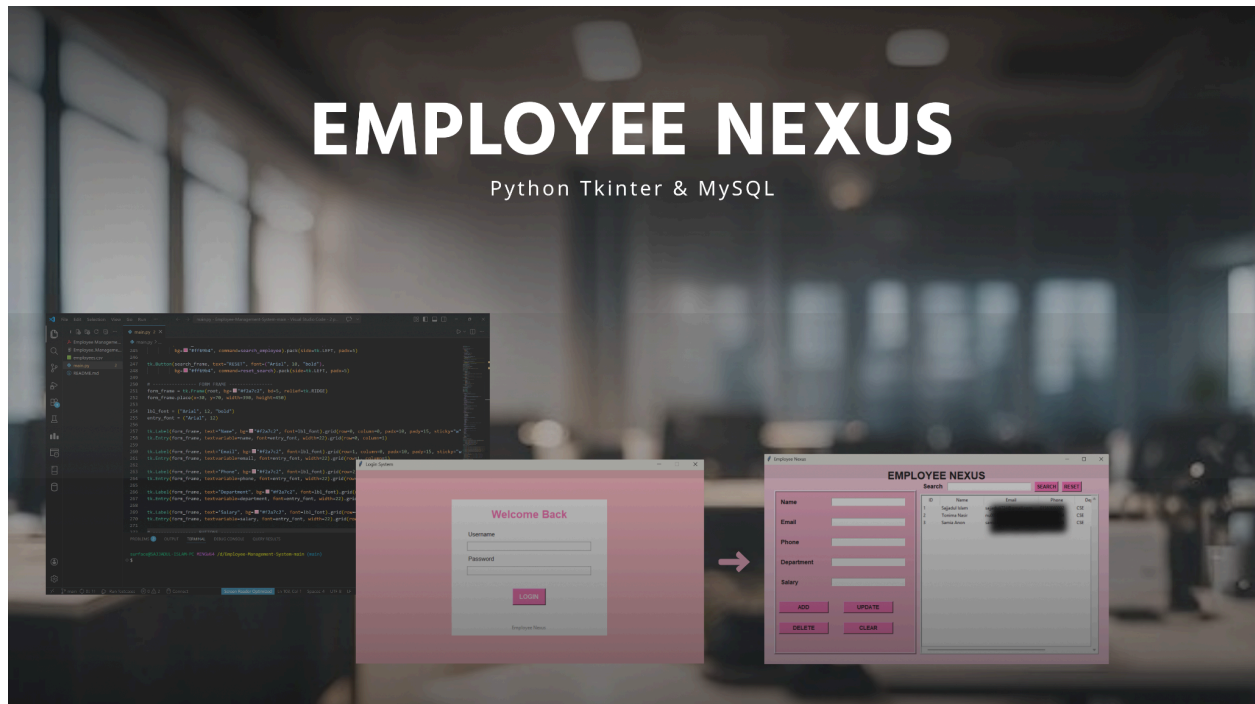
- Develop a user-friendly GUI application
- Store employee data in a database
- Perform CRUD operations
- Implement login security

1.3 Scope

This system can be used in small to medium organizations to manage employee data efficiently.

1.4 Organization of Report

This report describes system design, implementation, and future improvements.



Chapter 2: Background

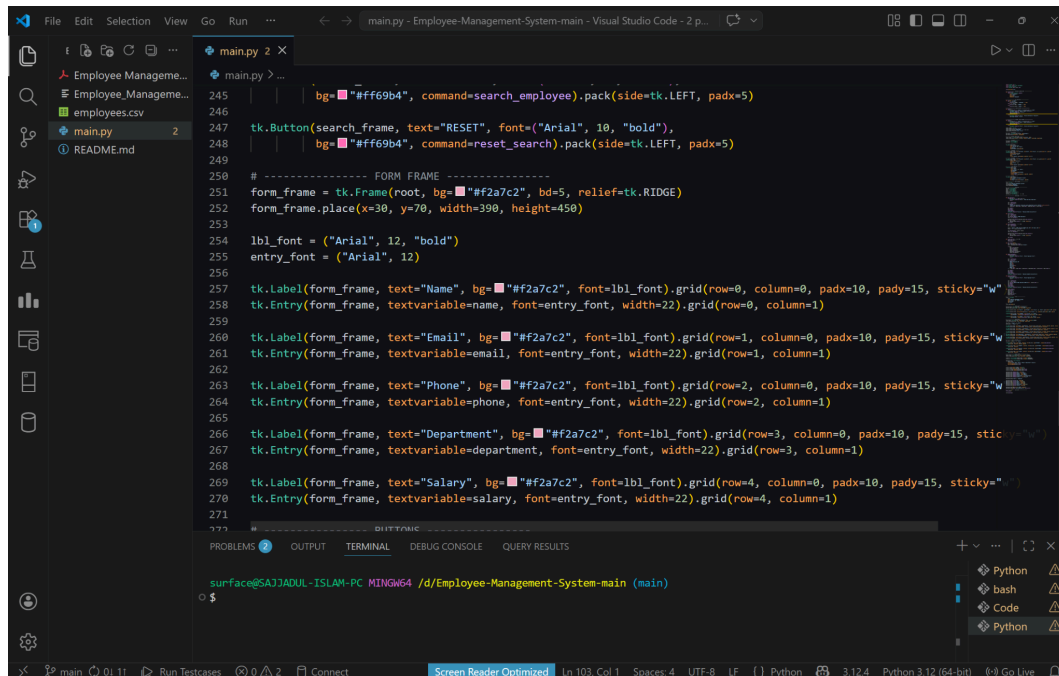
2.1 Existing System Analysis

Manual systems or Excel-based systems are commonly used but they lack automation and security.

2.2 Supporting Literature

Concepts used include:

- Database Management System (MySQL)
- GUI Development (Tkinter)
- Python Programming



```
main.py 2 X
main.py > ...
245 |         bg="#ff69b4", command=search_employee).pack(side=tk.LEFT, padx=5)
246 |
247 | tk.Button(search_frame, text="RESET", font=("Arial", 10, "bold"),
248 |         bg="#ff69b4", command=reset_search).pack(side=tk.LEFT, padx=5)
249 |
250 | # ----- FORM FRAME -----
251 | form_frame = tk.Frame(root, bg="#f2a7c2", bd=5, relief=tk.RIDGE)
252 | form_frame.place(x=30, y=70, width=390, height=450)
253 |
254 | lbl_font = ("Arial", 12, "bold")
255 | entry_font = ("Arial", 12)
256 |
257 | tk.Label(form_frame, text="Name", bg="#f2a7c2", font=lbl_font).grid(row=0, column=0, padx=10, pady=15, sticky="w")
258 | tk.Entry(form_frame, textvariable=name, font=entry_font, width=22).grid(row=0, column=1)
259 |
260 | tk.Label(form_frame, text="Email", bg="#f2a7c2", font=lbl_font).grid(row=1, column=0, padx=10, pady=15, sticky="w")
261 | tk.Entry(form_frame, textvariable=email, font=entry_font, width=22).grid(row=1, column=1)
262 |
263 | tk.Label(form_frame, text="Phone", bg="#f2a7c2", font=lbl_font).grid(row=2, column=0, padx=10, pady=15, sticky="w")
264 | tk.Entry(form_frame, textvariable=phone, font=entry_font, width=22).grid(row=2, column=1)
265 |
266 | tk.Label(form_frame, text="Department", bg="#f2a7c2", font=lbl_font).grid(row=3, column=0, padx=10, pady=15, sticky="w")
267 | tk.Entry(form_frame, textvariable=department, font=entry_font, width=22).grid(row=3, column=1)
268 |
269 | tk.Label(form_frame, text="Salary", bg="#f2a7c2", font=lbl_font).grid(row=4, column=0, padx=10, pady=15, sticky="w")
270 | tk.Entry(form_frame, textvariable=salary, font=entry_font, width=22).grid(row=4, column=1)
271 |
272 |
273 | # ----- BUTTONS -----
274 | btn_search = tk.Button(root, text="SEARCH", font=("Arial", 10, "bold"),
275 |         bg="#ff69b4", command=search_employee).pack(side=tk.LEFT, padx=5)
276 |
277 | btn_reset = tk.Button(root, text="RESET", font=("Arial", 10, "bold"),
278 |         bg="#ff69b4", command=reset_search).pack(side=tk.LEFT, padx=5)
279 |
280 | # ----- MAIN LOOP -----
281 | root.mainloop()
282 |
```

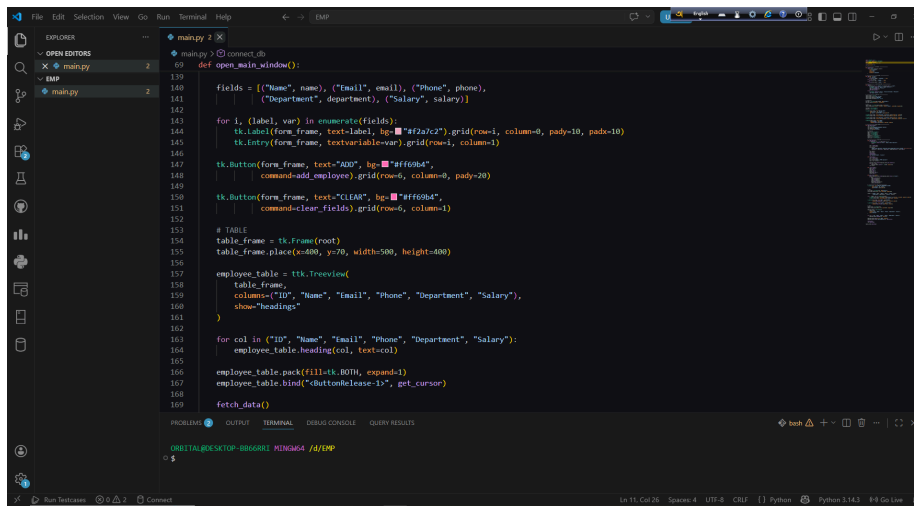
Chapter 3: System Analysis & Design

3.1 Technology & Tools

- Python
- Tkinter
- MySQL
- VS Code

3.2 Model & Diagram

- SDLC Model used
- Basic system flow includes Login → Dashboard → CRUD operations

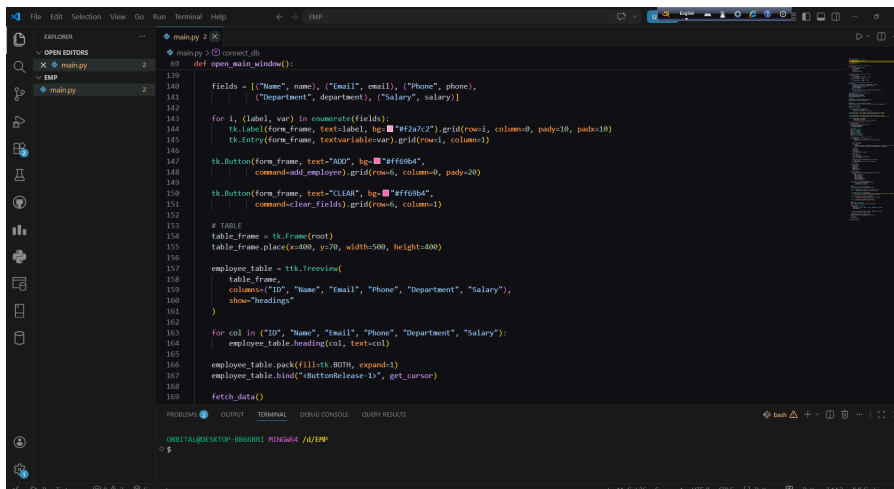


```
main.py 2 | connect_db
69 def open_main_window():
139
140
141 fields = [("Name", name), ("Email", email), ("Phone", phone),
142           ("Department", department), ("Salary", salary)]
143
144 for i, (label, var) in enumerate(fields):
145     tk.Label(form_frame, text=label, bg="#ff2a7c").grid(row=i, column=0, pady=10, padx=10)
146     tk.Entry(form_frame, textvariable=var).grid(row=i, column=1)
147
148 tk.Button(form_frame, text="ADD", bg="#ff69b4",
149           command=add_employee).grid(row=6, column=0, pady=20)
150
151 tk.Button(form_frame, text="CLEAR", bg="#ff69b4",
152           command=clear_fields).grid(row=6, column=1)
153
154 # TABLE
155 table_frame = tk.Frame(root)
156 table_frame.place(x=400, y=70, width=500, height=400)
157
158 employee_table = ttk.Treeview(
159     table_frame,
160     columns=("ID", "Name", "Email", "Phone", "Department", "Salary"),
161     show="headings"
162 )
163
164 for col in ("ID", "Name", "Email", "Phone", "Department", "Salary"):
165     employee_table.heading(col, text=col)
166
167 employee_table.pack(fill=tk.BOTH, expand=1)
168 employee_table.bind("<ButtonRelease-1>", get_cursor)
169
170 fetch_data()
```

3.2.5 Database Schema

Table: employees

- id (Primary Key)
- name
- email
- phone
- department
- salary



```
main.py 2 | connect_db
69 def open_main_window():
139
140
141 fields = [("Name", name), ("Email", email), ("Phone", phone),
142           ("Department", department), ("Salary", salary)]
143
144 for i, (label, var) in enumerate(fields):
145     tk.Label(form_frame, text=label, bg="#ff2a7c").grid(row=i, column=0, pady=10, padx=10)
146     tk.Entry(form_frame, textvariable=var).grid(row=i, column=1)
147
148 tk.Button(form_frame, text="ADD", bg="#ff69b4",
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164 for col in ("ID", "Name", "Email", "Phone", "Department", "Salary"):
165     employee_table.heading(col, text=col)
166
167 employee_table.pack(fill=tk.BOTH, expand=1)
168 employee_table.bind("<ButtonRelease-1>", get_cursor)
169
170 fetch_data()
```

Chapter 4: Implementation

4.1 Front-End Design

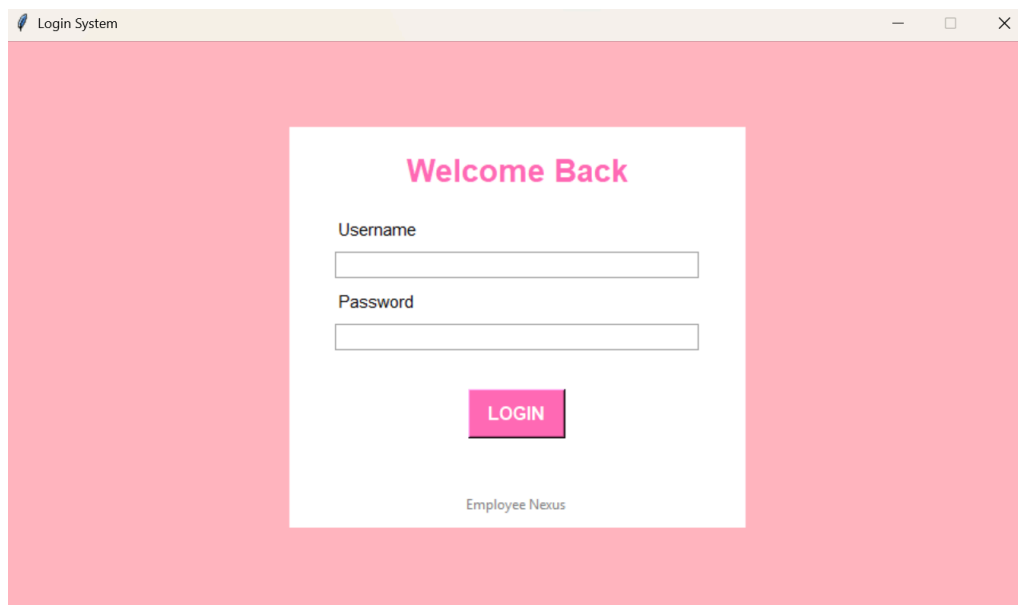
Tkinter GUI is used to design the login page and dashboard.

4.2 Back-End Design

MySQL database is used to store and retrieve data using Python.

4.3 Modules

- Login System
- Add Employee
- Update Employee
- Delete Employee
- Search Employee



Employee Management System

EMPLOYEE MANAGEMENT SYSTEM

Search **SEARCH** **RESET**

Name

Email

Phone

Department

Salary

ADD **UPDATE**

DELETE **CLEAR**

ID	Name	Email	Phone	Dej
1	Sajjadul Islam	sajjadul724@gmail.com	01560060092	CSE
2	Tonima Nasir	nu009012@gmail.com	01787295784	CSE
3	Samia Anon	samiaanon@gmail.com	01982217505	CSE

Chapter 5: User Manual

5.1 System Requirements

Hardware

- Minimum 4GB RAM
- Processor: Intel i3 or higher

Software

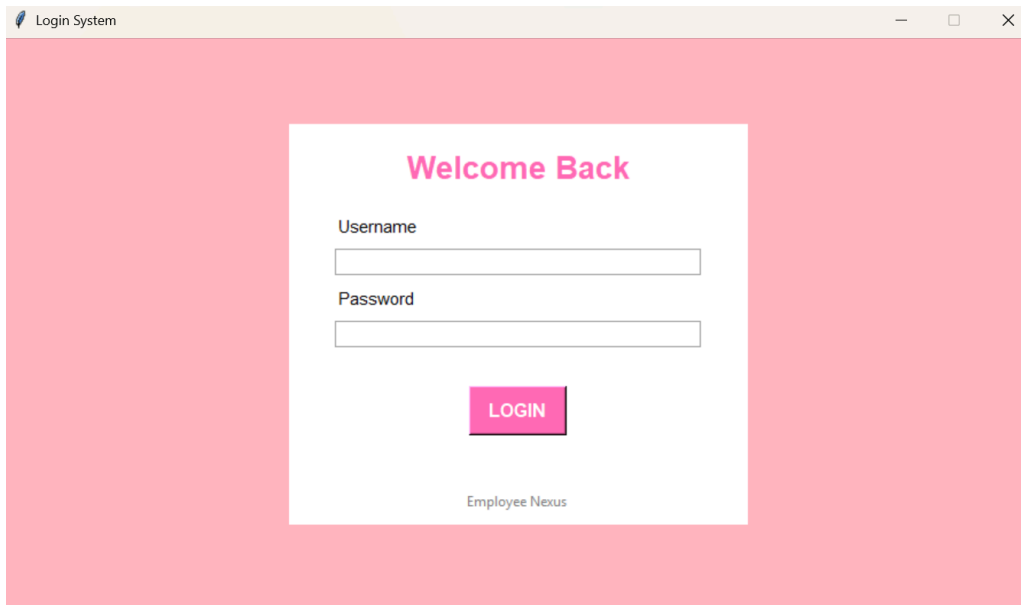
- Python 3.x
- MySQL Server
- VS Code

5.2 User Interface

Login Panel

Username: sajutonu

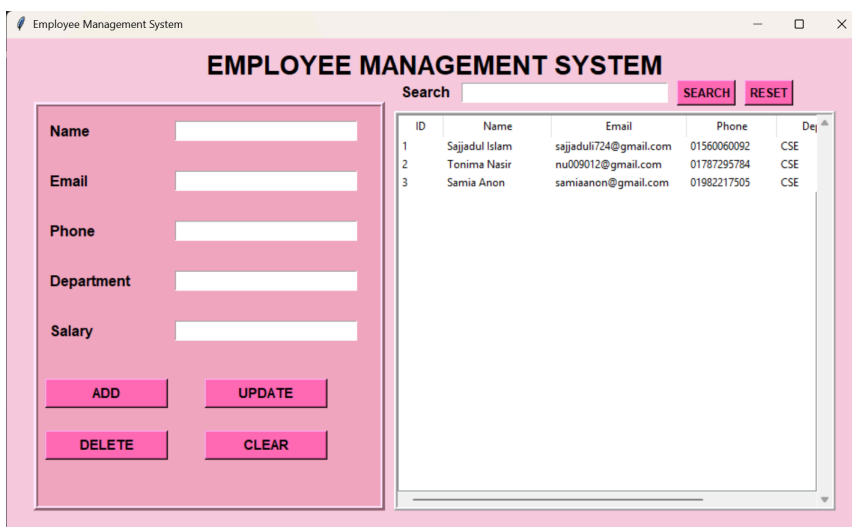
Password: 2508



The screenshot shows a web browser window titled "Login System". The background is a solid pink color. In the center, there is a white rectangular box containing the text "Welcome Back" in pink. Below this, there are two input fields: "Username" and "Password". Below the password field is a pink button labeled "LOGIN". At the bottom of the white box, the text "Employee Nexus" is displayed.

Dashboard

Users can manage employee data using buttons.



The screenshot shows a web browser window titled "Employee Management System". The background is a solid pink color. At the top, there is a header bar with the text "EMPLOYEE MANAGEMENT SYSTEM" in bold. Below the header, there is a search bar with the text "Search" and two buttons: "SEARCH" and "RESET". On the left side, there is a form with five input fields: "Name", "Email", "Phone", "Department", and "Salary". Below these fields are four buttons: "ADD", "UPDATE", "DELETE", and "CLEAR". On the right side, there is a table with five columns: "ID", "Name", "Email", "Phone", and "Dej". The table contains three rows of data.

ID	Name	Email	Phone	Dej
1	Sajedul Islam	sajadul724@gmail.com	01560060092	CSE
2	Tonima Nasir	nu009012@gmail.com	01787295784	CSE
3	Samia Anon	samiaanon@gmail.com	01982217505	CSE